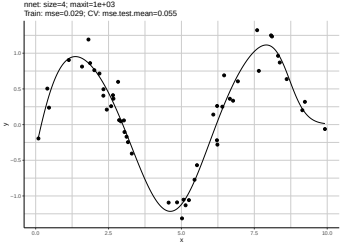
# Introduction to Machine Learning

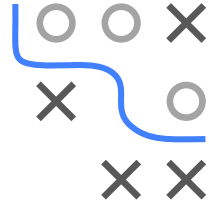
# Neural Networks
# Universal Approximation




## Learning goals

- Universal approximation theorem for one-hidden-layer neural networks
- The pros and cons of a low approximation error

# UNIVERSAL APPROXIMATION PROPERTY



**Theorem.** Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a continuous, non-constant, bounded, and monotonically increasing function. Let $C \subset \mathbb{R}^p$ be compact, and let $\mathcal{C}(C)$ denote the space of continuous functions $C \to \mathbb{R}$. Then, given a function $g \in \mathcal{C}(C)$ and an accuracy $\varepsilon > 0$, there exists a hidden layer size $m \in \mathbb{N}$ and a set of coefficients $W_j \in \mathbb{R}^p$, $u_j, b_j \in \mathbb{R}$ (for $j \in \{1, \ldots, m\}$), such that

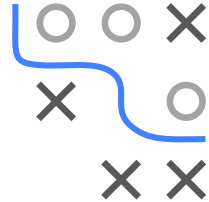
$$f : C \to \mathbb{R}\,; \quad f(\mathbf{x}) = \sum_{j=1}^{m} u_j \cdot \sigma\left(W_j^T \mathbf{x} + b_j\right)$$

is an $\varepsilon$-approximation of $g$, that is,

$$\|f - g\|_\infty := \max_{x \in C} |f(\mathbf{x}) - g(\mathbf{x})| < \varepsilon \ .$$

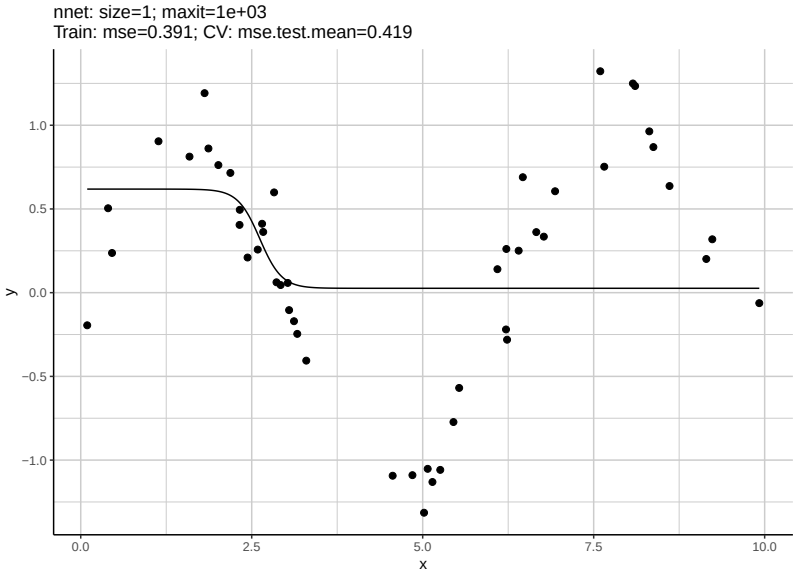
The theorem extends trivially to multiple outputs.

# EXAMPLE : REGRESSION/CLASSIFICATION

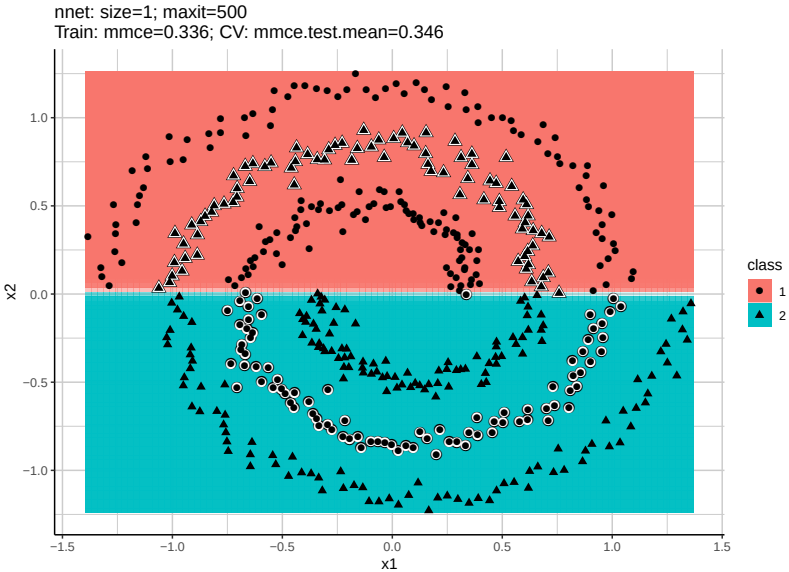


- Let's look at a few examples of the types of functions and decisions boundaries learnt by neural networks (with a **single** hidden layer) of various sizes.
- "size" here refers to the number of neurons in the hidden layer.
- The number of "iterations" in the following slides corresponds to the number of steps of the applied iterative optimization algorithm (stochastic gradient descent).

# REGRESSION EX.: 1000 TRAINING ITERATIONS



# CLASSIFICATION: 500 TRAINING ITERATIONS



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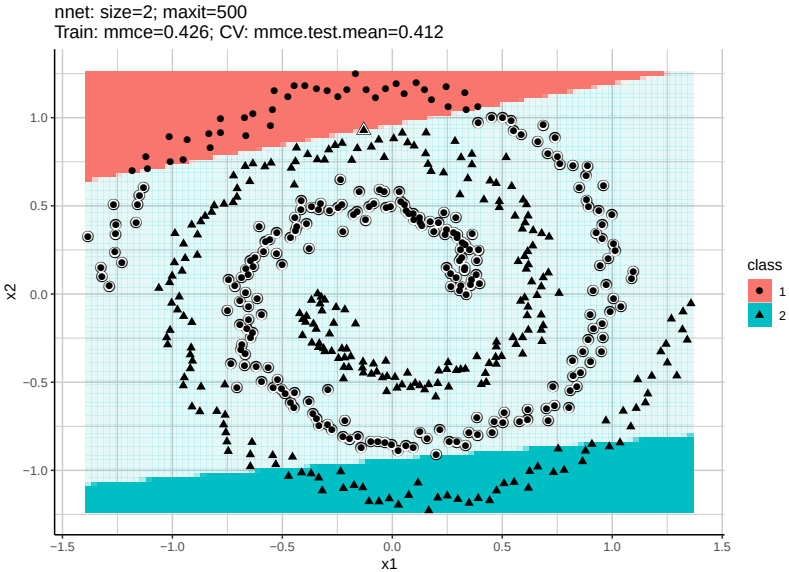

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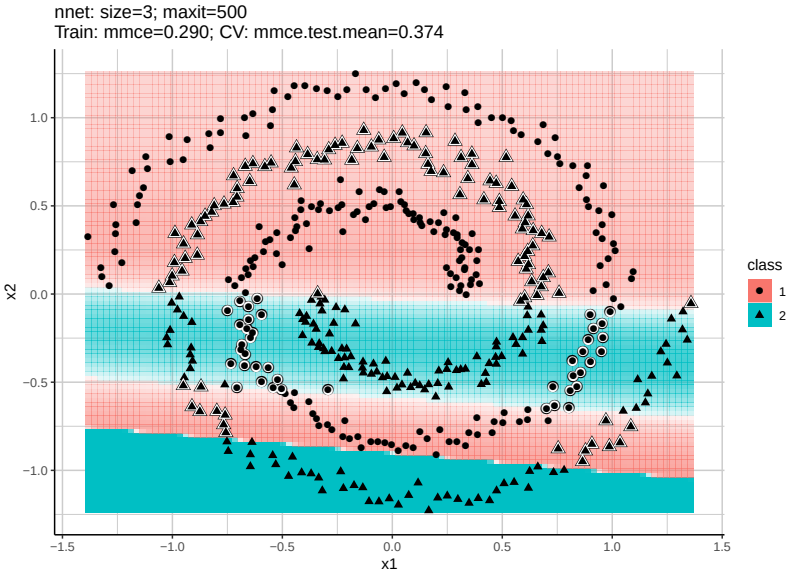

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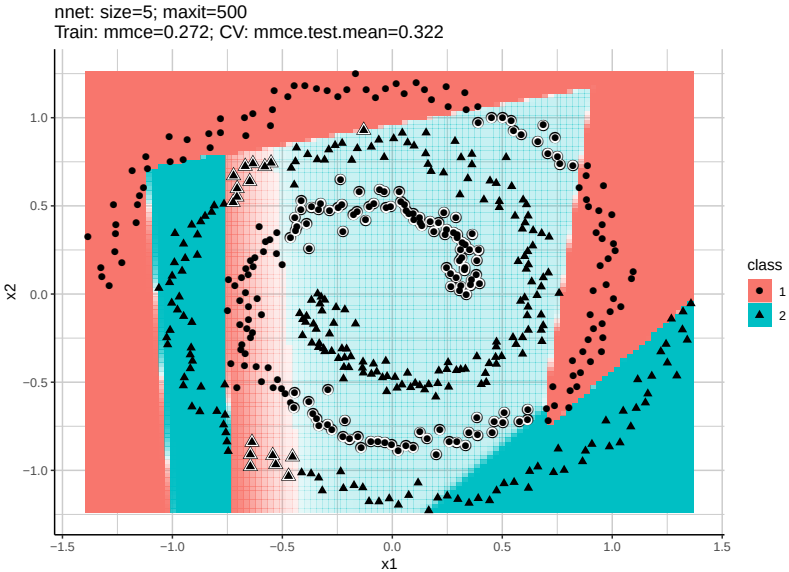

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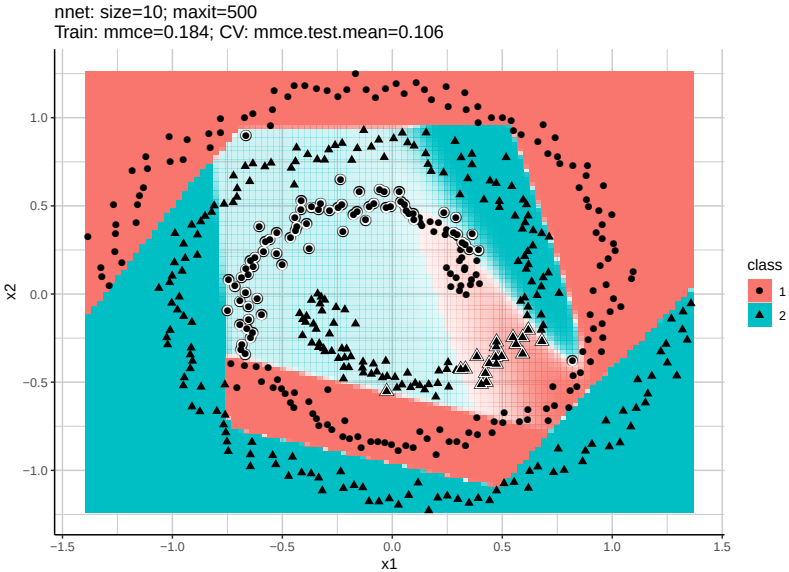

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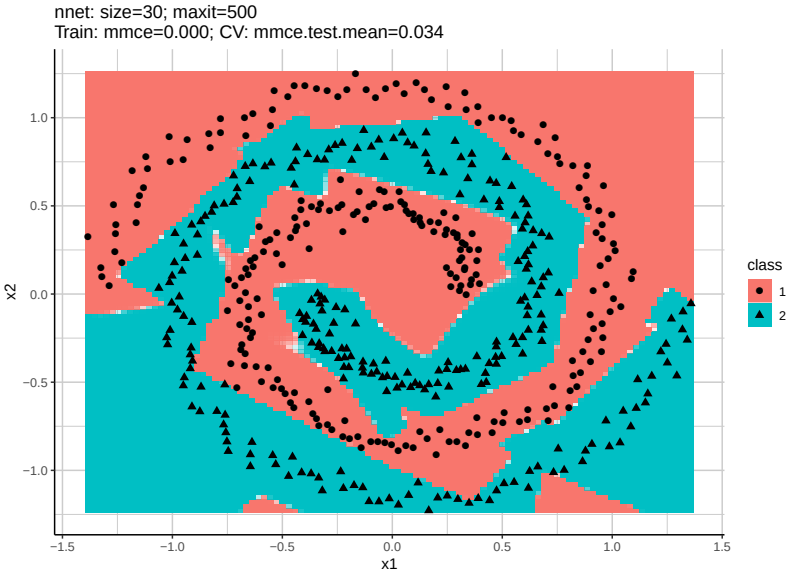
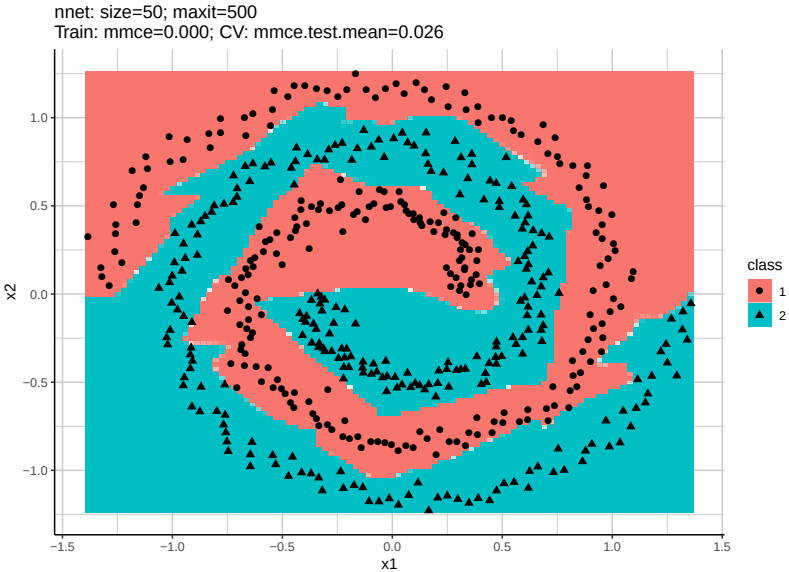
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